Runze Ma

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Education

Monash University Master of Artificial Intelligence 2024.07 – 2026.11
Huazhong University of Science and Technology B.Eng. in Automation 2020.09 – 2024.06

Publications

TRACE: Topology-Regularized Autoencoder for Interpretable Clinical ECG Diagnosis

- ***NeurIPS 2026****, CCF-A, Submitted,* **First Author (Co-Author)**
- We propose TRACE, a topology-regularized autoencoder that disentangles latent semantics into clinically meaningful subspaces via physiological constraints, achieving interpretable and robust ECG diagnosis.

LiteMedCoT-VL: Parameter-Efficient Adaptation for Medical Visual Question Answering *[link]*

- ***NLPCC 2026****, CCF-C, Submitted,* **First Author**
- We propose LiteMedCoT-VL, a 2B vision-language model achieves 64.9% on medical VQA by distilling chain-of-thought reasoning from a 235B teacher via LoRA, outperforming models with twice the parameters.

BEAT-Net: Injecting Biomimetic Spatio-Temporal Priors for Interpretable ECG Diagnosis *[link]*

- ***IEEE Journal of Biomedical and Health Informatics****, JCR Q1, IF 7.7, Under Review,* **First Author**
- We introduce BEAT-Net, a biomimetic framework mimicking cardiologist reasoning through QRS-centered tokenization and hierarchical spatio-temporal encoders for efficient, interpretable ECG diagnosis.

AEGIS-NET: Exploiting Physio-Kinetic Coupling Via Asymmetric Task-Aware Gating For Unified Biometrics, Activity Sensing, And Health Profiling *[link]*

- ***IEEE Transactions on Consumer Electronics****, JCR Q1, IF 9.9, Under Review,* **Second Author**
- We propose AEGIS-Net, a unified multi-modal framework for wearable devices that synergizes ECG, PPG, and IMU signals via an Asymmetric Task-Aware Gating mechanism.

Stacking-Enhanced Bagging Ensemble Learning for Breast Cancer Classification with CNN *[link]*

- ***ICEEM 2023***, **Second Author**
- We present BSECNN, a CNN-based ensemble using Bagging and Stacking for breast cancer classification.

REALM: Retrospective Encoder Alignment for LFP Modeling *[link]*

- ***Journal of Neural Engineering****, JCR Q2, IF 4, Under Review,* **Third Author**
- We propose REALM, a retrospective distillation framework that enables high-performance causal LFP decoding.

CPR: Causal Physiological Representation Learning for Robust ECG Analysis under Adversarial Perturbations *[link]*

- ***NeurIPS 2026****, CCF-A, Submitted,* **Fifth Author**
- We propose CPR, a robust diagnostic framework leveraging structural causal modeling to disentangle invariant pathological morphology from adversarial artifacts, achieving certified-level robustness and real-time inference efficiency.

Work Experience

NeuroML Lab at Tsinghua University *Research Assistant* 2026.05 – Present

- Developed and trained foundation models for biosignals.

Laboratory of Advanced Bioelectronics at SUAT *[link]* *Research Assistant* 2025.11 – Present

- Developed and trained multimodal pre-training models for biosignals.

Kingda Education *AI Technical Consultant* 2025.10 – 2026.06

- Delivered workplace AI training and provided technical consulting support for practical AI adoption.
- Designed and implemented task-oriented AI agents to streamline internal workflows.

Antalya Technology *[link]* *AI Product & Operations Lead* 2024.03 – Present

- Designed core prompts for **IELTS Master** *[link]*, constructing a structured CoT framework to improve stability.
- Extracted user data via SQL and built automated Power BI dashboards to drive product iteration.

Juxin Financial Consulting *Data Analysis Intern* 2024.11 – 2025.02

- Integrated OCR with LLM APIs to automate invoice processing, boosting reconciliation efficiency by 50%.

Project Experience

Wordmotion: One-Click Article-to-Video Pipeline with Remotion 2026.06 – Present

- Built a AI agent skill that turns articles into narrated Remotion videos with word-level subtitles.
- Designed modular TSX slide components with frame-precise edge-tts audio-subtitle sync and multi-platform support.

Recipe Generation: From RNNs to RAG Menu Designer 2026.04 – 2026.05

- Trained encoder-decoder RNNs with Bahdanau attention from scratch for ingredient-to-recipe generation.
- Fine-tuned T5 and GPT-2 with LoRA, and built a Menu Designer with RAG pipeline and LLM-as-Judge evaluation.

Multi-Agent RL for Autonomous Shuttle Delivery 2025.05 – 2025.06

- Designed a tabular Q-learning MARL system on a 5×5 grid with collision detection and hybrid reward shaping.
- All four agents achieved 100% delivery success rate in focal agent evaluation across 16 start configurations.

Pac-Man AI: Classical Search & Adversarial Planning 2025.03 – 2025.04

- Implemented A* search with domain-specific heuristics for single-goal, multi-corner, and full food collection pathfinding.
- Designed alpha-beta pruning with a handcrafted evaluation function for adversarial ghost avoidance.

Drone-Based Dual-Modal Vehicle Detection 2023.11 – 2024.06

- Improved YOLO with RGB-IR feature fusion to resolve detection failures in complex lighting and weather.
- Validated on DroneVehicle dataset and optimized quantization for real-time embedded deployment.

2023 Summer Ethics of AI Program at North Carolina State University *[link]* 2023.07 – 2023.08

- Surveyed AI ethical controversies and built a multi-dimensional assessment framework.
- Authored a report proposing targeted ethical guidelines, receiving positive mentor feedback.

Pain Intensity Estimation via Spatio-Temporal Networks 2023.02 – 2023.07

- Optimized feature extraction on UNBC-McMaster dataset by comparing multiple spatio-temporal architectures.
- Validated hybrid model advantages for non-invasive pain assessment via cross-validation.

Ensemble Learning with VAE for Breast Cancer Classification 2022.11 – 2023.02

- Proposed a Bagging & Stacking framework to address sample imbalance and improve reliability.
- Integrated VAE for data augmentation and feature selection to optimize classification performance.

Certificates & Honors

2025 Smart Wearable and Sports Health Technology Challenge - *National 1st Place* 2026.03
Scholarship for Scientific and Technological Innovation *[link]* - *1/30* 2022.09
Certificate of Professional Knowledge Assessment in Mathematical Modeling *[link]* 2022.08
Certificate of the University Artificial Intelligence Training Camp *[link]* 2022.08
MathorCup College Math Modeling Challenge *[link]* - *First Prize (Top 5%)* 2022.05
Mathematical Contest in Modeling *[link]* - *Honorable Mention (Top 20%)* 2022.02
Scholarship for Community Engagement *[link]* - *1/30* 2021.09

Skills

- **Programming:** Python > C = Matlab > Java > C++
- **Software:** SQL, Power BI, VibeCoding
- **Languages:** Chinese (Native); English (IELTS 6.5); Arabic (Basic Spelling)